

September, 2025

Global nickel investment horizon outlook Q3 2025



Executive summary

Nickel is facing a long period of over-supply. Such is the excess that no plausible improvement in our demand growth rate will reverse the situation. Perhaps tired of waiting for miracles to happen in the EU economy or for the west to establish a cohesive electrification strategy to boost nickel demand, market watchers get excited about anything from the supply side which could move the price needle. So far this year, whenever there is news from Indonesia which might curtail production, even for a short while, that is when prices have firmed.

Now accounting for 64% of global mined nickel production, and 65% of the smelter-to-intermediates total, it is understandable that only if there is a change in Indonesia will prices move out of their tight eight-month range of US\$14,800-15,200/t (US\$6.70-6.90/lb). We have lost metal from the west – Australia, mainly, and other areas too – but none of this makes a difference to the perception that Indonesia – and its Chinese paymasters – can merely open the faucet a little more to compensate for any loss elsewhere. When both countries can now produce most refined nickel products by type – and just as many desired intermediates – their degree of control over the nickel's fate is almost total.

To say that China has reached this position “by design” carries a tinge of bitterness from western commentators, knowing that fringe efforts to redress the imbalance, such as tariffs or environmental barriers like Europe's CBAM, are the only weapons in a depleted arsenal. But China has done to nickel no more or no less than what it has already done with other raw commodities, from iron ore to copper: seizing control. After being the west's supplier of low-cost goods for two decades, China has the dollars to invest in areas which ensure it will not be dependent on the west for the goods needed to guarantee its industrial future. So rather than bitterness, it is with envy that western commentators should look upon this long-term view from Beijing. Everything indeed has been “by design” – set in stone by a long-term view which western investors cannot, or do not want to, afford.

After a whirlwind of expenditure in Class II facilities in the early 2000s, we have long noted that western investors pulled back from new nickel prospects, and for good reason. But, oddly right now, there appears to be renewed interest in Africa's sulphide resources. Most developments are still some years away – ironically, the one which might make fastest progress is Chinese-owned – but should they gain ground, they might enter production at the turn of this decade when some existing mines will be looking at remediation rather than a productive future. Whether they will make a conventional concentrate to feed existing smelter capacity to maintain the long-held status quo of current producers, or whether they enter into new relationships to make products more directly suited for the west's hoped-for electrification, is not yet clear. But, in light of China's current and likely future dominance in everything nickel, it is an interesting and, in some ways, surprising development which merits more than the occasional eye.

Something else which should merit closer attention takes us back to demand – and back to China. In July 2024 Beijing initiated a discussion about the problems being faced by several sectors of the economy. In short, too many producers and service providers of all goods – from electric vehicles (EVs) to food delivery companies – were operating at slim to zero margins because of a price war which was threatening the stability of entire business sectors and the nation's financial strength. From that initial discussion and subsequent Central Committee gatherings in Q1 and Q2 2025 emerged “guidelines” to end harmful and unfair competition by, for example, ending subsidies or tax incentives, or by raising quality standards. “Value, not volume” will be the new mantra when these guidelines get ratified by Beijing and provincial capitals from early 2026.

What we expect to see ultimately from this strategy is fewer names in every game, including nickel-related products. The range of EV makers in China might shrink to a fraction of the number we see today, and such contraction will follow down through all suppliers in the chain. Whenever the need for this contraction ends, it might not necessarily mean “less” nickel, cobalt or lithium being required if the names which do remain are indeed able to satisfy real demand for the EV models and chemistries of the day. But anti-involution measures might mean that the path forward is not as straight as we expect today. Indeed, the contraction already witnessed in China's stainless steel melting sector gives us advance warning of what might come in the EV and battery supply chain sector over the next five years. The long-term trend might be up, but it will not be all plain sailing – or “driving”.

Investment landscape

Nickel oversupply – what happens next?

The global nickel market remains one that is all about China and Chinese investments in Indonesia. There are facets of interest in other parts of the world – for example the recent interest in using Indonesian NPI in European stainless steelmaking – but these days it is really only China that moves the dial.

The extraordinary growth in global nickel supply that equates to a CAGR of 10% since 2021, to 3.9 Mt in 2025, is due to the rapid wave of expansion in Chinese-funded NPI and nickel intermediates production in Indonesia, the latter primarily to supply raw materials for the lynchpin of the energy transition – batteries for EVs.

Ironically, one might think that with annual growth of 5-7.5% in the past three years, global demand has been doing well. But even at those healthy rates demand has fallen far behind supply, leading to a cumulative surplus of 650 kt. That will increase by another 370 kt in 2025 and the accumulation will not end until the late in the decade. Almost all the growth in demand through this period was in China – both in stainless steel and battery chemicals, while markets elsewhere struggled, undermined by declining economies and sentiment in the face of global geopolitics and conflict in Ukraine and the Middle East. Further confusion and uncertainty came this year with the wave of tariffs imposed by the Trump administration in the US, and that will likely extend into 2026.

Even in China, markets were not as strong as the ramp up in raw materials production implied. Growth in stainless steel was mostly tied to the ramp up of new, large facilities of Tsingshan and Tisco at a time when the market could not support all that additional stainless product supply. Other companies have taken the hit, low demand and low prices forcing them to cut production or exit the stainless segment all together. In the battery stream it is slightly different. The matte and HPAL plants in Indonesia have sustained run-rates exceeding demand because their intermediates can also be processed all the way through to Class I metal destined for LME warehouses – the Chinese are producing metal for stock. But the nickel sulphate market is relatively balanced as a result.

So, essentially the global nickel market is suffering under the double whammy of strong oversupply and weak demand. Class I nickel is in substantial oversupply – reflected by the increase in LME stocks from 37 kt in mid-2023 to this month's level of 228 kt. Class II nickel is also in oversupply because of NPI over-production in Indonesia, much of which is taking place at Weda Bay (IWIP).

For this to change either demand has to improve – and for this we look to China – or supply has to be reined in – and again we look to China, or at least Chinese investments in Indonesia.

Stronger markets will eventually return to Europe, US and the rest of Asia, but while that will boost global demand numbers it is a stronger China that really makes the difference. Chinese stainless will continue to increase and there are still several projects in the pipeline to support that. However, this could be countered by ongoing consolidation, something that may additionally be influenced by the Chinese government's policy of anti-involution. A further downside risk is that increases in primary nickel uptake could be undermined by increased use of scrap, as scrap becomes more available and the need to both recycle more and reduce carbon emissions becomes more desirable.

China's battery precursor (pCAM) segment is about to surge ahead as new plants come online later this year and into 2026. This lies behind our forecast 14-15% a year rise in nickel demand from this industry over rest of this decade. This contributes towards a narrowing of our annual surpluses through that period. But of course, that is already factored into our analysis so to make further material difference to the oversupply the growth in demand would need to be even greater than the above figure, which some may already feel is too high.

Oversupply and low nickel prices have made life difficult for the world's nickel producers forcing many of them to close, such as the European FeNi pool and BHP's Nickel West in Australia. Others remain at risk – Doniambo in New Caledonia and Ambatovy in Madagascar are two we have often cited, although recent developments at both have reduced that risk – while Terrafame will soon be suspending nickel sulphate production for an unspecified period. While these losses have sometimes triggered sentiment-driven but temporary spikes in the nickel price, they have made little impact on market oversupply. Further cuts or closures in the mature economies will only have a similarly muted impact.

A change in Indonesian production strategies is really what needs to happen. But even there it has not been plain sailing for all producers. Bankruptcy proceedings for Jiangsu Delong reduced output at the company's Gunbuster NPI asset. Other companies too have had to cut back as margins were squeezed by rising ore prices, fluctuating NPI prices and weak demand from the stainless mills. That the Bahodopi project is now an HPAL rather than NPI further affirms the challenges of the NPI market in Indonesia. At the same time, retarded ramp up at the sulphate projects of QMB and Huaxiang point to the weaknesses in sulphate demand.

But while cuts are clearly taking place, the problem is that there always seems to be other producers ready to make up for those losses – ramp ups at Weda Bay, Halmahera Jaya and Huabao, and a resurgent Virtue Dragon. Weda Bay alone will produce more than 500 kt from this year, or 12% of *global* nickel supply.

What can change this? The Indonesian government continues to make announcements that at least implies a desire to control production rates more effectively. The recent changes to benchmark prices and the increase in royalties have helped to squeeze margins a little more. The government's strategy for mining quotas – RKABs – is also key. There was much speculation a year ago

on the reported intent of the government to restrict quotas which we interpreted as a way to sustain higher grade nickel resources, limiting their extraction while new resources could be developed. Expectations that quotas would be cut in 2025 supported a rally in nickel prices but in the end the announced quota for 2025 was *larger* than in 2024. Maybe the trend will be downward in 2026?

The added intention to reduce mining quotas from three years to one year may give the government more control over who mines where, but it would also promote greater uncertainty in ore supply and greater potential for disruption through delays in issuing permits, both of which would engender greater volatility in the nickel price. But whatever the length of the permit, unless genuine cuts are made to ore supply tonnages, the oversupply in the market will remain.

The idea that Indonesia produces “dirty nickel” and that the world should seek to use “green nickel” from somewhere else instead is flawed because in the end there is no real alternative. Such has been the growth in nickel production in Indonesia and the lack of capacity addition anywhere else in the world, if you need nickel some will have to come from Indonesia – especially since Russian nickel is also off the agenda these days. That said, the Indonesian nickel industry has wide environmental issues that need addressing, from deforestation to land run-off, emissions and encroachment on indigenous groups that continue to incite protests.

In the past four months such issues have led the Indonesian government to cancel mining permits for companies in Raja Ampat, Kabaena and Gebe. This month a pocket of land was “seized” from Weda Bay Nickel but this too was not of a size large enough to make any difference to nickel production. However, the temporary suspension on 18 September of 190 mining companies that failed to provide reclamation guarantees highlights that greater stringency is coming, although the 36 nickel mines affected in Sulawesi and North Maluku are small and relatively insignificant so in nickel terms the government is still only “tinkering round the edges”. With Indonesia advancing its next stage of nickel development in battery materials and the EV supply chain, ESG standards will be more closely scrutinised by more people worldwide than the predominantly Chinese audience it has currently.

Market outlook and price

Medium-term forecasts

Little has changed in the past 12 months to correct the extent of oversupply that began in 2022. With the surplus in 2025 alone heading towards 400 kt we now anticipate a cumulative excess of 1.51 Mt for the 2022-2029 period, before the balance switches to negative in 2030.

And yet nickel prices stubbornly remain around US\$15,000/t. In some respects, this makes sense as the historic relationship between costs and price are that annual prices approach the 70th percentile in times of weak markets and oversupply. US\$15,000/t is around that price point and as that relationship is also a guide to where prices may go in the years ahead, the ongoing decline in nickel prices levels out. So, while spot values may dip lower, month averages will edge down toward US\$14,000/t and will stay there until 2028.

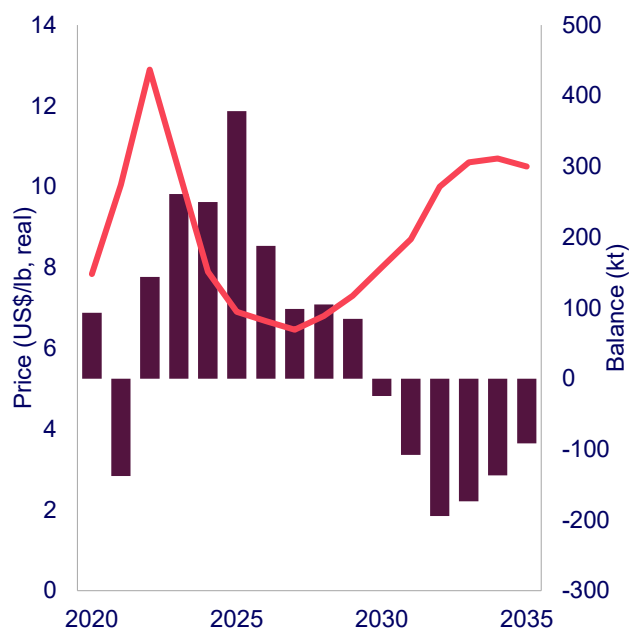
At the same time, looking at a single global nickel balance no longer reveals enough about the present market dynamics – a market that these days is really three markets in one: Class I nickel in substantial surplus; Class II nickel overproducing in 2025 but returning to balance; and nickel sulphate broadly balanced and getting tighter from 2026.

Even though EV sales continue to increase, the demand for nickel in battery materials is not so strong, largely due to the large share of the battery market now taken by non-nickel LFP (Li-iron phosphate). However, Chinese ternary pCAM production increased significantly in Q3 and with a lot of new capacity about to come on stream in the next 12 months there could be a sizeable step up in nickel consumption in this sector.

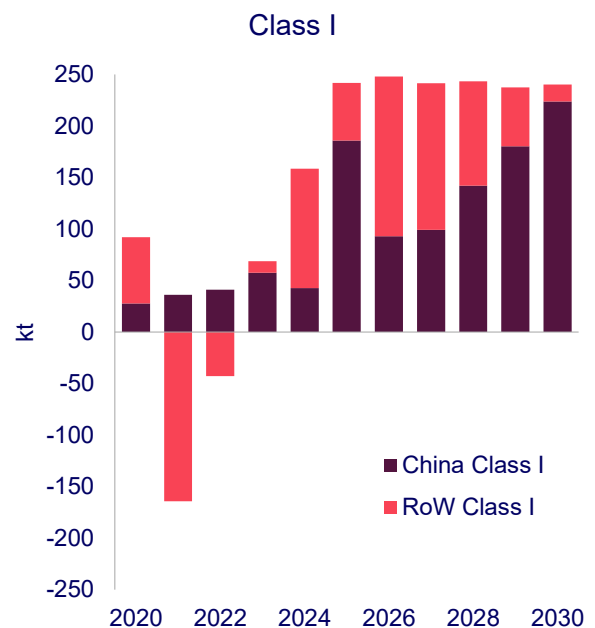
Given operational capacity, the sulphate market surplus could be much higher were it not for the option of many Chinese producers to instead process their intermediates all the way to Class I metal. For those with approval, this metal can then be placed in LME warehouses – the market of last resort – and it is this practise that is creating the large surpluses in Class I supply. Chinese companies are producing Class I metal for stock, and that excess is clearly visible as the rise in LME stock by around 200 kt in the past two years. But with the sulphate market tightening, more sulphate will be needed in the years ahead so this will cap or reduce Class I production levels in the future.

Class II nickel jumped into unprecedented surplus in 2022 due to surging NPI production in Indonesia as new capacity came on stream. This continued into 2023 before slowing last year, to a level roughly in line with the NPI still residing in Indonesia waiting to be shipped to China. However, over-production has returned in 2025 as still more new capacity ramps up with a potential surplus by year end of 160 kt. We assume that in the future the Class II market will revert to convention and rebalance – after all, there is no market of last resort for NPI or FeNi. In doing so, the market can counter any deficits by drawing on the existing overhang. Alternatively, Chinese mills can seek additional nickel units by raising their use of stainless steel scrap.

Nickel balances by product type



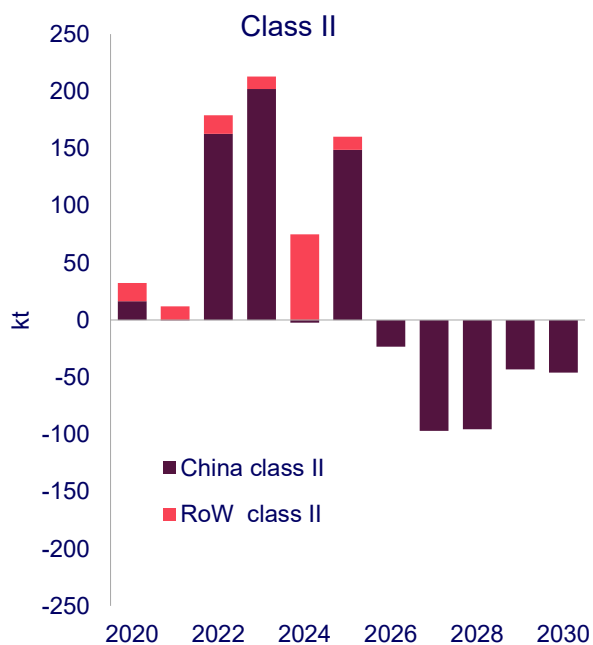
Source: Wood Mackenzie



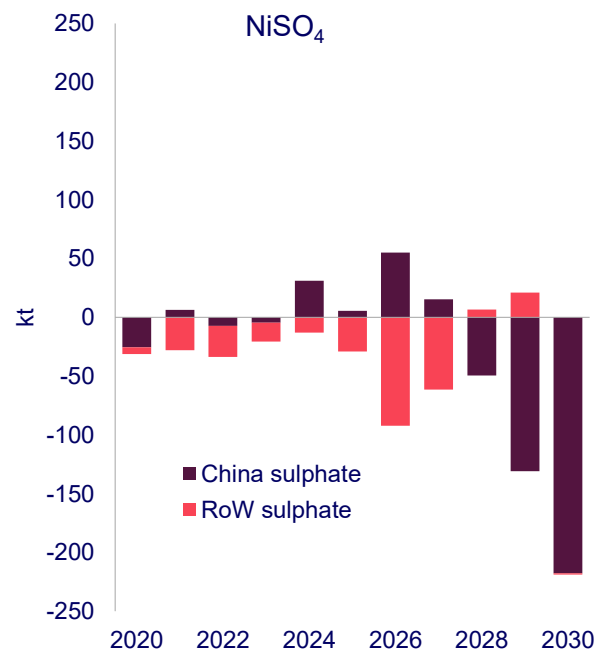
Source: Wood Mackenzie

Source: Wood Mackenzie

Nickel balances by product type



Source: Wood Mackenzie



Source: Wood Mackenzie

Such are the projections for Indonesian production that growth in global nickel demand of 8.8% in 2026 followed by healthy rates of around 5% a year in 2027 and 2028 makes little impact on the continuous accumulation of still more stock. And that is before we consider the risks to demand going forward.

Firstly, the figures are leveraged to China where we still see stainless steel expanding but more importantly have a surge in demand from battery chemicals as aforementioned capacity starts to come on stream later this year. But the risks are that both markets continue to disappoint: stainless prices stay low, while Tsingshan and Tisco dominance combined with governmental anti-involution

drives consolidation amongst their competitors; further penetration of LFP batteries undermines the need for all the new ternary pCAM capacity.

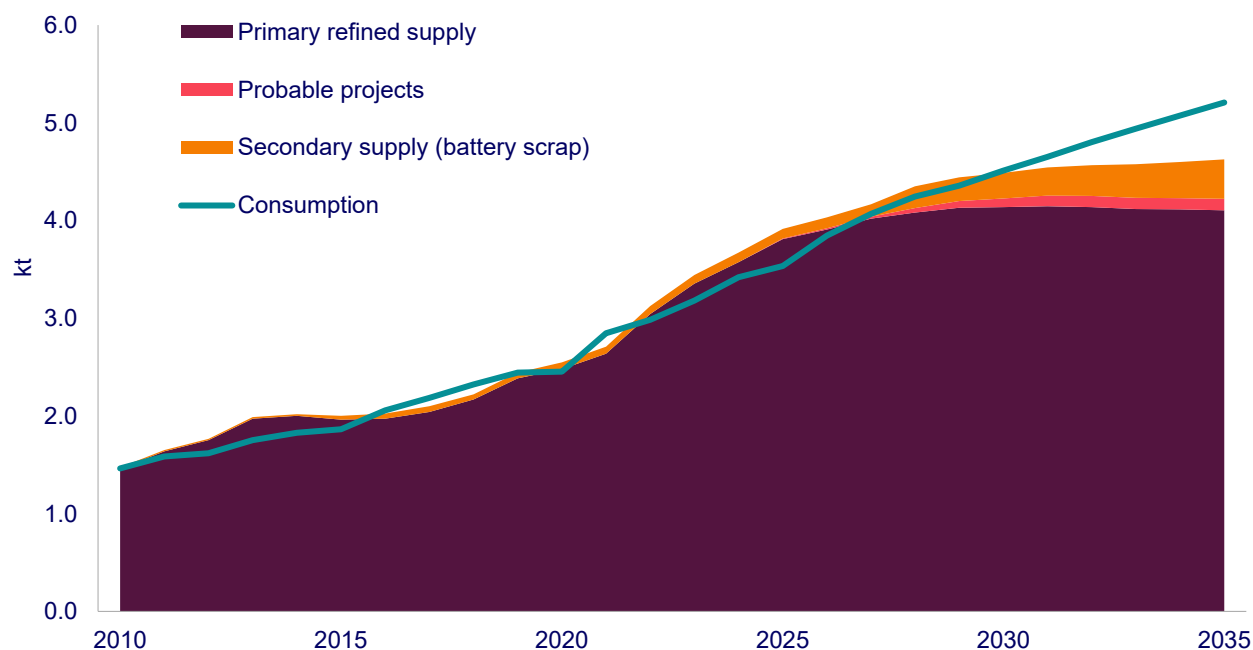
Elsewhere, the stainless industries in Indonesia and India will expand, the former due to new capacity (from Jindal), the latter due to stronger demand from key segments such as infrastructure. In battery materials, new pCAM capacity in Indonesia and Morocco as well as Korean investments in other countries will boost the new capacity starting up in China. But for Europe and the US it all depends on better economies and lower uncertainty which are both suffering under the pressure of geopolitics, conflict and the off-and-on announcements of trade tariffs. For sure, improvement will come but the question is when? One should also be wary of percentage growth figures that may look strong, but in reality are coming off the low bases of prior years.

Whether one's view on demand is positive or negative, the impact on the global nickel balance is small compared to a meaningful adjustment to production strategy in Indonesia. Cuts there are realistically the only thing likely to move the dial and we summarise how this could begin in the Investment Landscape section above.

So, with the nickel market still oversupplied and with prices still trading in a narrow range around US\$15,000/t (US\$6.80/lb) our forecast is only slightly changed since the Q2 Strategic Planning Outlook. Oversupply will remain until 2029-2030 before a period of substantial deficit takes hold in 2031-2035. As a result, this year's forecast average nickel price of US\$15,210 (US\$6.90/lb) will drop slightly in 2026 and 2027 before starting to improve, finally rising above US\$16,000/t, in real terms, in 2029. Projected shortage in the early 2030s will push prices up to a peak of between US\$23,000/t and US\$24,000/t (US\$10.40-10.90/lb) in 2033-2034 before settling at our long-term price of US\$20,285/t (US\$9.20/lb).

By 2032, the nickel market will need 40 kt of new nickel supply from somewhere – either from possible projects or as yet unidentified resources - to facilitate a reasonable long-term balance. This requirement increases to 490 kt in 2035, 1.19 Mt in 2040 and 1.46 Mt in 2050.

Mid-term nickel supply-demand balance



Source: Wood Mackenzie

Nickel sulphate premiums

Consistent price disparities exist between Chinese and overseas nickel sulphate markets, rooted in different primary feedstocks. China's nickel sulphate tends to be traded at a discount due to its reliance on lower-cost mixed hydroxide precipitate (MHP), while overseas production predominantly dissolves higher-cost nickel metal.

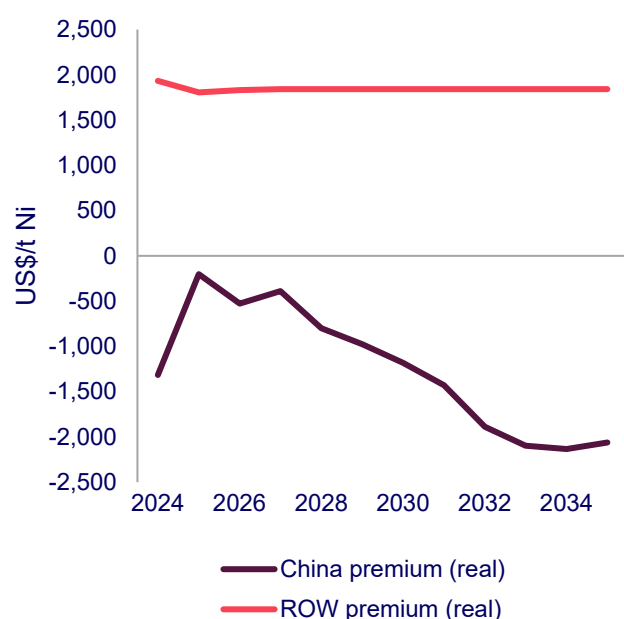
China's 2025 nickel sulphate prices are expected to average a US\$200/t Ni discount to LME nickel. Since Q2, nickel sulphate's discount has narrowed significantly, turning to a premium in Q3. This shift was driven by tight MHP supply following Indonesia's Q2 floods and recovering battery demand in Q3. The Democratic Republic of Congo's cobalt export ban also boosted cobalt payability, further supporting MHP payability and sulphate prices.

Despite the relatively strong performance in the recent quarter, we expect annual prices to remain discounted over the forecast period. The broader EV industry shift toward lithium iron phosphate (LFP) chemistries has weighed on demand for nickel-based battery materials, and we do not expect this trend to reverse course in the near term. Besides, expanding Indonesian MHP production will likely limit future price surges.

Although China's domestic nickel sulphate balance shows a slight deficit, this could be easily eliminated as producers switch between sulphate and full plate nickel (Class I) production based on market conditions. We project the discount will widen slightly to US\$526/t in 2026 and then expand further to around US\$2,000/t post-2033.

Outside China, nickel sulphate maintains consistent premiums. Rest-of-world production, predominantly via higher-cost briquette dissolution, commands a premium of around US\$1,800/t nickel throughout the forecast period. In contrast, 2025 export data reveals notable regional variations: Japanese nickel sulphate averaged a premium of US\$4,500/t nickel, Finnish material US\$1,200/t, and Korean sulphate US\$1,900/t.

Nickel sulphate premiums



Source: the Argus Media group; Wood Mackenzie

	Nominal premium (US\$/t Ni)	Nominal price (US\$/t gross)	Real premium (US\$/t Ni)	Real price (US\$/t gross)
2024	-505	3620	-523	3747
2025	338	3452	338	3452
2026	119	3371	116	3292
2027	343	3372	329	3234
2028	109	3563	103	3351
2029	78	3892	72	3589
2030	-23	4326	-21	3911
2031	-168	4767	-149	4225
2032	-491	5524	-427	4800
2033	-652	5946	-555	5065
2034	-691	6117	-577	5109
2035	-652	6132	-534	5020

Source: the Argus Media group; Global Trade Tracker; Wood Mackenzie

Risks to forecast

Short-term upside

The Indonesian government takes more concerted action to suspend nickel mines and other facilities that are inadequately permitted or thought to be contravening environmental and other ESG regulations.

Chinese demand surges in Q4: increased stainless melt production in Q3 continues over the rest of the year; new pCAM capacity starts on time and ramps up quickly to extend the rally in pCAM production that began in Q3.

Short-term downside

Anticipated strengthening of demand reverses, particularly in the battery chemicals segment – new pCAM capacity is deferred – leading to greater production of Class I metal, which in turn shows up as a surge in LME nickel stock.

To “beat the tariffs” or “beat the subsidy removal deadline”, a lot of purchasing especially at the retail level has been front-loaded this year, especially in the US. From EVs to seasonal holiday gifts, many market watchers believe Q4 sales will be poor because they’ve already been snapped up. The decline in shipping rates since May on conventionally busy routes between Asia and North America and low tonnes of vessels being scrapped already point to poor Q4 expectations in trade circles.

Medium-term upside

A step change in Indonesian nickel supply across intermediates and finished nickel products that reflects continuing oversupply being unsustainable; production rates are cut back and new capacity is deferred.

Anti-involution in China could drive consolidation amongst nickel producers, especially NPI, and reduce nickel output.

General economic improvement, geopolitical uncertainty wanes, conflict in Ukraine and the Middle East ends and market confidence returns.

Medium-term downside

Anti-involution could equally disrupt demand: stainless steel is already entering a new round of consolidation because of the ramp up of new capacity when the market did not need it; the rapid expansion in EVs could also be retarded which would in turn cut demand for nickel in battery pCAMs.

Nickel consumption is undermined by further increases in the share of LFP in China's battery mix.

Uncertainty over global tariff regimes.

Demand

Overview

As in 2024, global economies continue to feel the effect of conflicts in Ukraine and the Middle East, enhancing uncertainty and keeping sentiment risk averse. Regional markets for stainless steel and battery materials, while variable, remain weaker than hoped for at the start of the year.

China remains the main driver of 3.3% growth in global nickel consumption in 2025, to 3.54 Mt, although neither its stainless nor battery chemicals industries had a convincing first half. However, both segments picked up significantly in Q3. Elsewhere, markets are decidedly weaker, especially Europe's stainless market, which will contract for the third year in the past four. Import tariffs instigated by the Trump administration appear to have supported short-term growth in the US stainless segment.

Looking ahead, China will still be the main driver of growth in global nickel demand owing to its dominance in both stainless steel and battery materials. Support will come from Indonesia, where new stainless and battery pCAM capacity is due to start up later this year, and India, where government investment in infrastructure will support the stainless industry. South Korea will also increase its battery chemicals output.

For other mature nations, as little new melting capacity is planned, output will be kept steady by the constraint of existing capabilities. Any movement back toward prior highs is dependent on economic recovery and stronger markets. Battery materials production, however, remains small. Japan has made little effort to expand its capabilities while all the recent promising efforts in the US and Europe to develop battery chemicals supply chains seem to be unravelling.

On this basis, a surge in Chinese pCAM capacity will push up growth in global nickel consumption to 9% in 2026, to 3.85 Mt, and then by a CAGR of 3-4% a year, to 4.5 Mt in 2030 and 5.2 Mt in 2035.

Stainless steel will remain the main consumer of primary nickel over the mid-term, accounting for 63% of the market in 2025 but falling to less than 51% by 2035, as the share taken by battery pCAMs rises to more than 30% or more.

Regional nickel consumption

									CAGR	Change kt	CAGR
	2025	2026	2027	2028	2030	2035	2040	2050	'25-2035	'25-2035	'25-2050
Europe	281	289	299	307	333	364	369	377	2.6%	83	1.2%
United States	122	122	125	130	141	161	173	208	2.9%	40	2.2%
Japan	176	176	179	182	187	197	201	221	1.1%	21	0.9%
China	2256	2444	2558	2687	2831	3305	3925	4422	3.9%	1049	2.7%
Indonesia	393	469	530	549	566	605	631	682	4.4%	212	2.2%
South Korea	109	131	148	155	184	241	248	252	8.2%	132	3.4%
Others	198	213	229	233	269	334	369	434	5.4%	136	3.2%
Global Total	3535	3846	4068	4243	4510	5207	5916	6597	4.0%	1673	2.5%
Source: Wood Mackenzie											

Changes since the last quarter

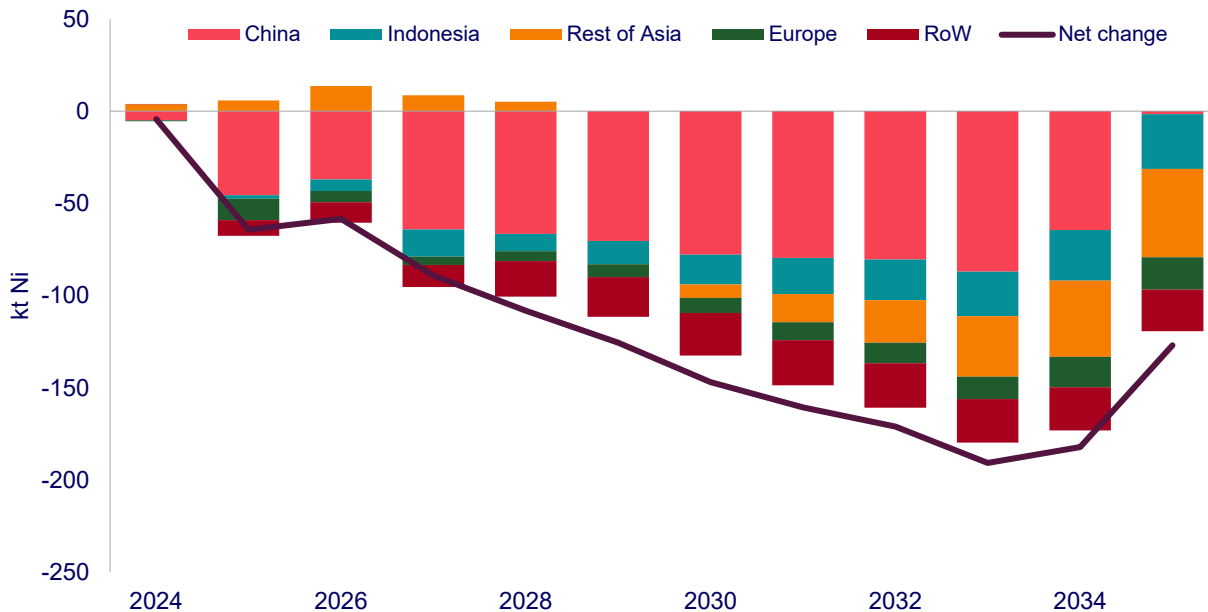
Changes to our forecasts since the last quarter mostly relate to stainless steel in the near-term and battery pCAMs in the mid-term.

In stainless, reduced production in China and Indonesia related to production cuts in June and July that were triggered by the weak Chinese market. Weak markets elsewhere also led to downward revisions to our forecasts for South Korea and Taiwan, as well as in Europe for 2025. These changes have a knock-on impact of 2026, 2027 and beyond.

In contrast, more substantial decreases in our mid-term outlook for nickel in pCAMs is due to reduced forecasts for nickel uptake in pCAM as the strengthening preference for LFP in China slows not only the expansion of ternary pCAM capacity

there, but also the evolution towards higher nickel battery types. In the rest of the world progress with complete battery material supply chain has stalled in comparison with a couple of years ago.

Changes in nickel consumption since Q2



Source: Wood Mackenzie

Nickel in stainless steel

Africa

Acerinox's Columbus mill is the only stainless steel producer in Africa. Historically, the mill has depended on export markets in Europe and Asia, including the supply of hot-rolled stainless to the company's former Bahru cold-rolling mill in Malaysia. However, challenging markets and the rise of Indonesian stainless meant that progressively lower tonnages were sent to Bahru as Columbus lost out to Indonesia, and this ultimately led to the sale of that plant in 2023. With recent European markets also struggling Columbus started to produce carbon alloy steel from 2020. As the rise in carbon steel helped to keep utilisation rates up it also led to stainless melt production falling, from a pre-covid level of around 550-600 ktpa to 200-300 ktpa subsequently.

Going forward, Acerinox is changing its strategy to focus more on southern African markets so that export volumes shrink from around 70% of the plant's business to only 30%. Product mix will also change, with electrical steel being added to mild steel and carbon alloy while stainless steel production is focused more ferritic grades. With neither electrical steel (which can be referred to as "stainless") or ferritic stainless containing any nickel the strategy will severely reduce or end the plant's exposure to volatile nickel prices. We tentatively forecast long-term stainless output of around 300 ktpa with zero requirement for primary nickel.

Asia

Asia will remain the powerhouse behind global growth in stainless steel production, chiefly through further increases in capacity in China, Indonesia and India, the latter helped by recent government measures to support infrastructure projects and Jindal's investments in Indonesia. In contrast, there is little expectation for growth elsewhere: production in Japan and South Korea will remain steady at around 2.3 Mtpa each, with the risk on the downside; Taiwan has been using Indonesian slab and hot-rolled stainless since 2017 rather than melt its own stainless, so melt production has halved since then and will remain steady at about 700 ktpa.

Regional stainless melt production will increase by 1.2% in 2025, to 54,5 Mt, with growth in China and India being offset by contraction in Indonesia, Japan and Taiwan. A stronger China and India, as well as new capacity in

Indonesia will mean stronger growth of 6% in 2026, to 57.8 Mt, before that slows over the following years, reaching 70 Mt in 2030 and 72 Mt in 2035.

That China and Indonesia produce more stainless than is needed by China alone (demand in Indonesia is very small) means that the Chinese continue to find ways for their low cost stainless to penetrate foreign markets. The reaction by producers in those markets has been to lobby governments to put tariffs – mostly antidumping or countervailing duties – on stainless originating from China and more recently, Indonesia. Asian markets have not been immune to this but as new duties spread so Chinese semis flowed to other countries for processing, initially to Taiwan, Malaysia and Vietnam, and most recently Turkey – in an effort to circumvent existing duties.

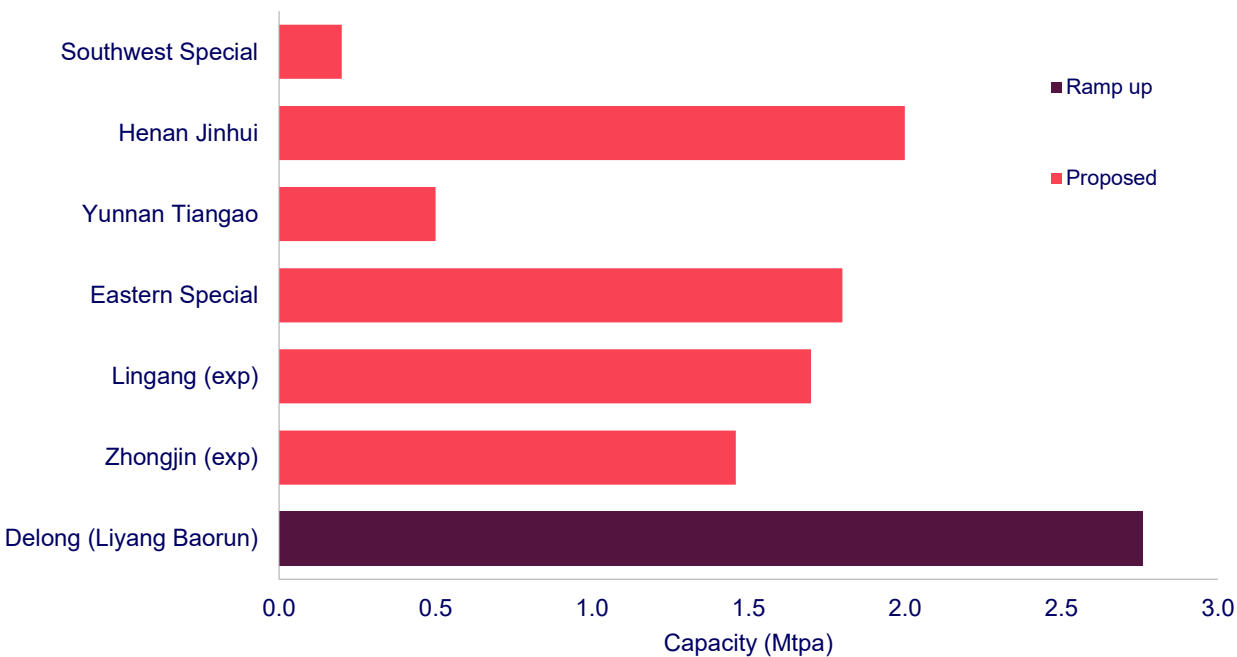
While the latter three countries have no stainless melting of their own, the growth in imports of Chinese stainless product has affected the local businesses of Posco (VST in Vietnam and Assan in Turkey) and Acerinox (Bahru in Malaysia), which in turn has impacted the melting activities of both companies at home, in Korea and South Africa and Europe.

This trade battle is now 10 years old and shows no sign of coming to an end. Indeed, Chinese offshoring continues with the latest cold rolling concern of Yongjin, a close ally of Tsingshan that processes in Vietnam much of the latter company’s Indonesian output, recently breaking ground on a new plant in Thailand.

China

Growth in China’s stainless melt production has slowed to less than 3% this year and will exceed 40 Mt. Stronger growth in 2023 and 2024 was largely driven by the ramp up of new capacity by Tsingshan and Tisco which were counter to market strength. While the domestic market may yet recover some of its lost fervour in 2026 and beyond, future growth to 49 Mt in 2030 and 53 Mt in 2035 also depends on further capacity expansion. But while there are projects in the pipeline, there is a general lack of clarity on timelines and status.

The pipeline for Chinese stainless melt shop capacity



Source: Wood Mackenzie

In some respects, it is debatable whether there is any real need for more capacity. Although national melt shop utilisation is a healthy 86% there are still more than 6 Mt pa of spare melting capacity available, much of that operated by Jiangsu Delong, one of China’s largest stainless producers, albeit one that has struggled recently. Delong has only emerged from bankruptcy proceedings this year so it is only now that the company is starting to fill more than 3 Mtpa of new capacity across two plants in China, albeit slowly. But the Mintal and Bishi stainless projects (more than 2 Mtpa between them) that we have known about for a long time were both shelved in the past year.

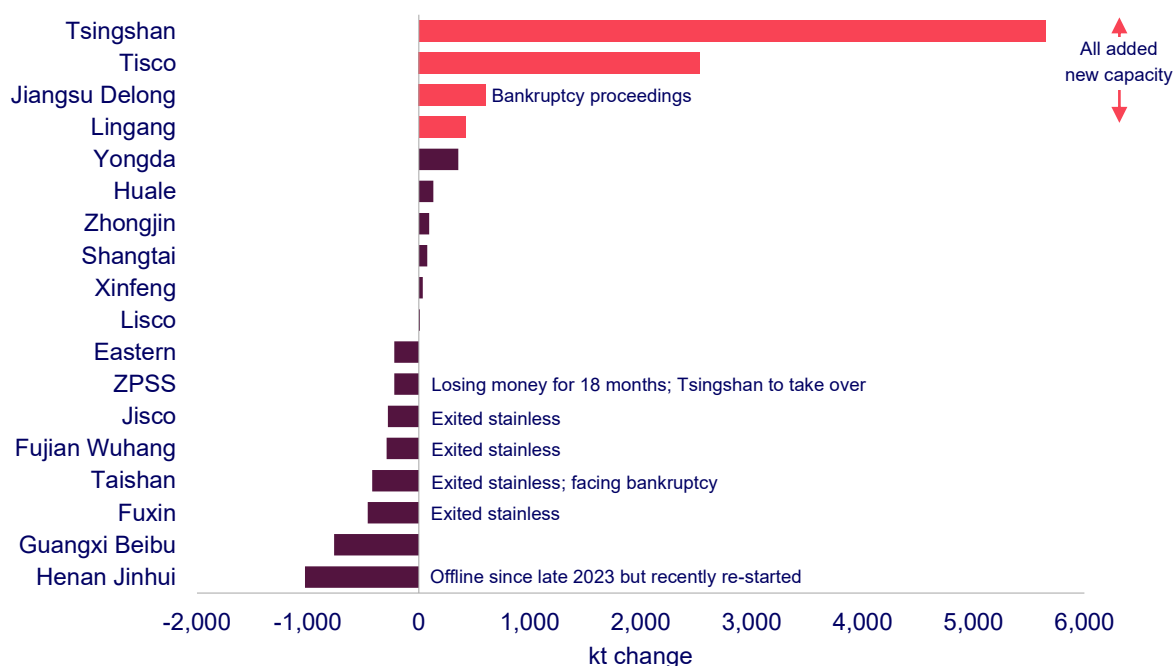
One of the factors contributing towards Delong’s recent struggles is the dominance of Tsingshan and Tisco, and their

efforts to bring online large new plants in deference to the strength of the market. The success of these ramp ups took market share from many of their competitors. Given the weak market conditions and low stainless prices, shrinking margins forced many of those competitors to reduce production or in some cases exit the market all together, as happened to Jisco, Wuhang, Taishan and Fuxin. The latest casualty is ZPSS, the Posco subsidiary, which has lost money continuously over the past 18 months and is in the process of being sold to Tsingshan.

So Tsingshan gets larger while its competitors are forced out. This is a trend that will drive further consolidation in the years ahead. For those stainless plants that resist and manage to keep operating, their product portfolios may expand to include higher value specialties: Shandong Lingang (part of Shanyang Gp.) completed trial production of its 100 ktpa “high-end stainless seamless pipe project” which may transition from 300s only to 400s, duplex and perhaps even nickel alloys in the future; Zhongjin, which may not now follow through on its expansion plans, recently added ultra-thin 200s stainless coil to its product list; and Henan Jinhui, only recently restarted after more than a year offline, is still working on plans for a 2 Mtpa stainless industrial park for high quality and high precision sheet, including wide plate, that could be ready by 2030.

It will be interesting to see how the government’s policy of anti-involution – to counter unsustainable practises that lead to a “race to the bottom”, such as expanding capacity to create the appearance of market domination - affects this trend. Anti-involution will also drive consolidation but it should prize value over volume so it will favour those companies trying to expand into new niches with value-added products. But perhaps it will not be so good for the capacity-addition strategy of Tsingshan and Tsico?

Changes in Chinese stainless production since 2022, by plant



Source: Wood Mackenzie; ZLJ Steel

With growth in Chinese stainless production slowing over the second half of the decade, growth in primary nickel uptake (1.51 Mt in 2025) will also decelerate. However, the use of primary nickel – as NPI, FeNi and pure metal – could fall quicker than that depending on the use of stainless steel scrap. In recent years China’s austenitic scrap ratio has been fixed at around 20%, largely because NPI is abundantly available. But, as stainless production increases NPI supply will tighten, which will prompt greater use of scrap. Furthermore, the importance of lowering carbon emissions and pressure to have product lines that are ESG-friendly should steadily increase preferences for scrap going forward.

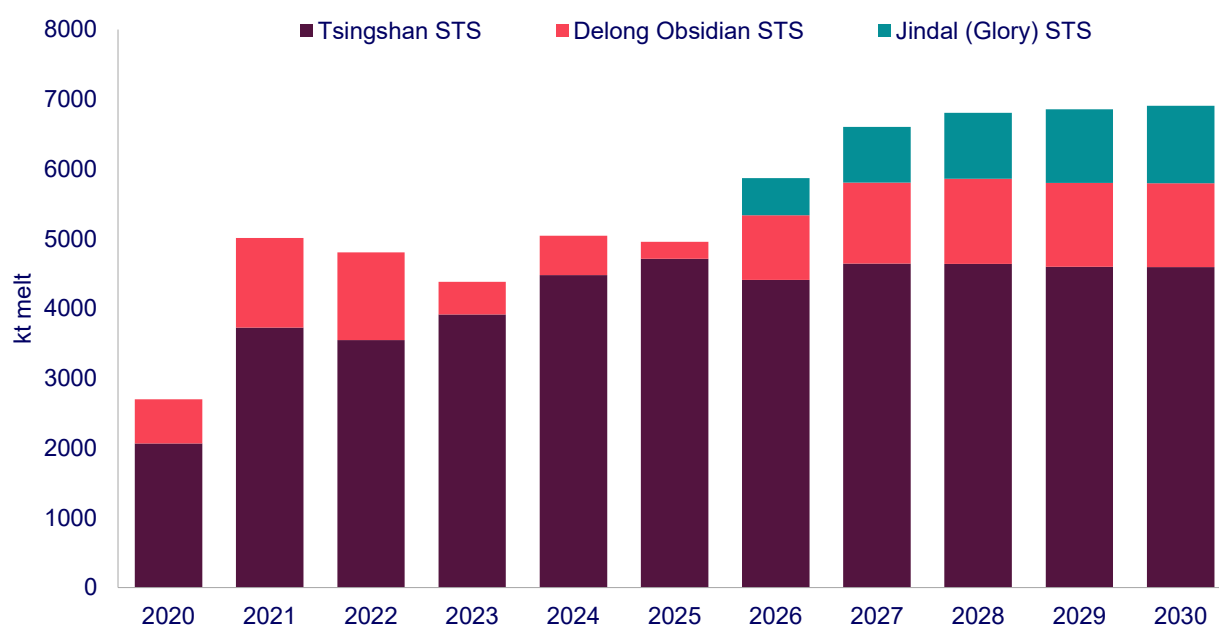
By 2035, we expect Chinese scrap ratio to exceed 30% thereby limiting nickel consumption to 1.73 Mt in that year, the equivalent of a CAGR of 1.4% over the 2025-2035 period. Market uncertainties mean we maintain a scrap ratio of 32% over the long-term, but clearly it could go higher. Indeed, a rise in Chinese scrap ratio to something more akin to a “western” norm, could cut primary nickel uptake by something of the order of 1 Mt by 2050.

Indonesia

Indonesian production is steady at around 5 Mt largely supported by Tsingshan Morowali at about 4 Mt with the Delong-Xiamen Xiangyu Obsidian plant accounting for the rest. Obsidian was forced to close again this year as Xiamen worked to sort out Delong's debt and because of the mid-year weakness in the Chinese market. However, the plant re-started in August and we assume it will slowly ramp up from the low run rates characteristic of the past two years.

Jindal's new Glory Metal stainless complex is due to start up by the end of 2025 primarily to send stainless semis to India. This will provide the main boost to the Indonesian total in the near term, supported by any increases achieved at Obsidian. On that basis, we see Indonesian stainless melt output increasing to 6.9 Mt in 2030 before levelling off at 7.2-7.3 Mtpa from 2035.

Outlook for Indonesian stainless melt production, by company



Source: Wood Mackenzie; ZLJ Steel

India

India's stainless steel sector is currently balancing growth in domestic demand with persistent external pressures from imports. The import trajectory over the past four years underscores both the challenge and the opportunity for Indian stainless mills.

Cold-rolled stainless steel imports rose from 219 kt in 2021 to 437 kt in 2022, peaking at nearly 600 kt in 2024. This put significant pressure on domestic producers and depressed utilisation rates across India's 7.0-7.5 Mtpa installed capacity, which continues to operate at around only 60%.

Annualised imports for 2025 suggest a moderation to around 500 kt, yet volumes remain high enough to displace domestic output. With anti-dumping petitions against Chinese and Vietnamese (processed and transhipped Chinese product) flat products already filed and a stainless-specific policy under discussion, imports should continue to ease over the medium term. If duties are implemented effectively, imports could fall further, enabling domestic mills to expand production more rapidly in response to growing consumption.

Stainless production is expected to increase only modestly this year as imports continue to absorb part of the market, by 1% to 4.37 Mt, compared with 4.32 Mt in 2024. This limited growth reflects the reality that low-cost Chinese origin imports remain highly competitive despite recent weakening. However, the medium-term picture is optimistic. For 2026, production is forecast to rise by 4.2%, to 4.55 Mt, supported by rising domestic consumption and policy enforcement, although there is upside to this given ISSDA's forecast of 8% growth in stainless consumption. By 2027, output is expected to climb to 4.85 Mt, up 6.6%, as safeguard measures, infrastructure demand, and improved domestic capacity utilisation converge. Beyond that, solid growth may be sustained to the end of the decade if the ISSDA target of 12 Mtpa

melting capacity by 2030, rising to 20 Mtpa by 2047 is realised.

Jindal Stainless is leading current investments at Jajpur and Hisar and is evaluating a 4 Mtpa greenfield complex in Maharashtra. Shyam Metalics, Viraj Steel, BRG Steel and SAIL are also expanding, supported by both private and public investment.

The drivers for the growth of stainless steel consumption include railway expansion, promotion of stainless rebar in coastal construction, water pipelines and the emerging use of stainless in green hydrogen storage. According to ISSDA, rising per capita consumption of stainless steel in India now stands at 3.2 kg compared with 6 kg in developed markets, which also leaves scope for structural long-term growth.

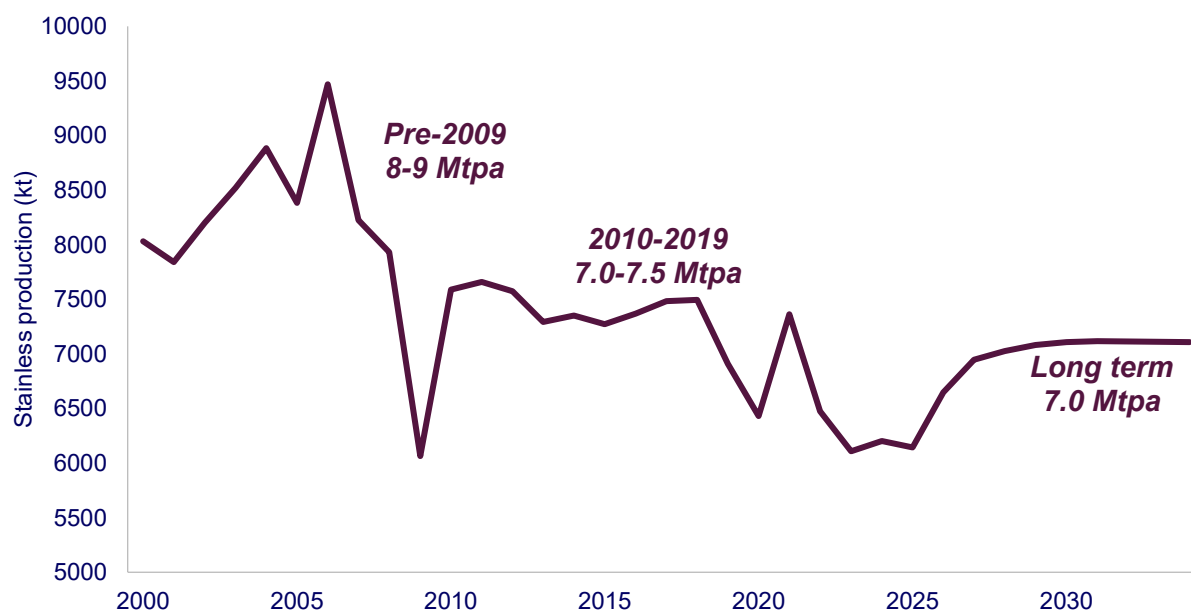
Also, recent measures, such as the removal of FeNi duty, the extension of zero-duty stainless scrap imports, and BIS certification requirements, have eased input costs and reinforced quality standards. A stainless-specific policy under discussion could further support the sector, recognising its outsized revenue contribution relative to crude steel.

Europe

The European stainless steel market has been in a state of contraction since 2021, when production hit a peak of 7.4 Mt. Since then, weak demand and low prices have forced output lower, reaching 6.2 Mt in 2024, and no more than that is expected in 2025. The latest results from the major producers - Acerinox, Aperam, and Outokumpu -all demonstrate that while they are managing costs and operations effectively, the market fundamentals remain weak.

European melt production will fall by 1% in 2025, to 6.15 Mt, a significant change from the earlier, more optimistic prediction of 5% growth. However, our long-term outlook remains positive, assuming that economies improve and uncertainty is replaced by renewed confidence. We anticipate recovery in 2026, with 8% growth taking melt production of 6.65 Mt, and that continuing in 2027 to 6.95 Mt. While such percentage figures may appear unrealistically strong to some, they still leave regional output tonnages well below the 2021 peak and pre-pandemic productivity levels.

European stainless production - steadies at a new low level of 7 Mtpa



Source: Wood Mackenzie

Imports remain a key challenge for the European market. In 2022, European cold-rolled stainless imports peaked at 1.3 Mt before falling back in 2023 and 2024. This year, they have increased sharply again, to 482 kt for January to July (annualised as 826 kt), potentially up 30% on 2024.

This rise is primarily driven by rerouted Asian stainless steel from China and Indonesia, which is also finding its way into Europe via countries like Taiwan, Vietnam, and Turkey to circumvent existing duties. This influx has put significant pressure on European mills, eroding their market share and compressing margins.

The Carbon Border Adjustment Mechanism (CBAM), which begins in earnest from 1 January 1, 2026 may help to rebalance the market. While CBAM is designed to prevent carbon leakage into and out of the region, recently it has also been viewed as protectionist. However, as we have seen with other duties, there is a risk imports will surge in late 2025 as customers and traders attempt to "beat the tax." Once CBAM is enforced, however, the comparatively high-carbon-emission stainless from China and Indonesia will be less competitive, potentially allowing European producers to reclaim some of the market share lost to imports over the past several years.

At the same time, raw material dynamics continue to evolve. European mills have relied on imported Indonesian NPI over the past two years as stainless scrap availability tightened and prices became untenable. The substitution of NPI for scrap raised regional nickel consumption even as stainless melt volumes declined but we expect some rebalancing in 2026, when scrap tightness should ease. In similar fashion, some European producers are importing stainless slab from Indonesia this year in preference to melting and casting their own semis. This practice is driven by the high costs, especially of energy, in Europe and may stop if energy prices ease or stainless demand returns allowing stainless prices to rally. Equally, the use of Indonesian slab may also be precluded once CBAM fees become payable.

Competition from low-cost Asian imports is also nudging European producers away from commodity stainless, such as grade 304, where competition is most acute. Instead, they are channeling investment into higher-value, high-performance alloys and specialty products, and edging further into the nickel alloys space.

Aperam has long been part of this segment through its Imphy subsidiary but recently expanded into the US alloys segment with the acquisition of Universal Stainless and Alloys. Acerinox recently entered the nickel alloys space through acquisitions of VDM (Germany) and Haynes (US). Even Outokumpu is pivoting towards high-nickel alloys and low-CO₂ materials at its Avesta melt shop, while using its EVOLVE program to concentrate its stainless operations at the cost-efficient Tornio complex in Finland.

These strategic moves signal that the European market could become increasingly polarised. The commodity segment may be increasingly fed by low-cost imports from Asia, while the high-performance segment will be dominated by European and North American players. CBAM taking effect and producers continuing to implement diversification into new niches should both support the optimistic outlook we forecast for Europe's nickel consumption.

North America

At 1.11 Mt, the first half of this year was the highest six-month period of stainless melt production in the US since April-September 2021 (1.21 Mt). This has put the national forecast on track to breach 2.0 Mt for the first time since 2022. Whether this upward trend will continue or not is clouded by uncertainties in both the domestic US market and the international trade climate.

At home, there are as many negative views as there are positive regarding how the economy will play out, with the former predicting a return to stagflation and the latter expecting home production to rally because of import tariffs. What we do know is that once the section 232 tariffs became enforced in 2018, both exports and imports of stainless products initially declined; subsequently, imports resumed in spite of the tariffs, yet exports did not pick up, and it was this drop in export-led demand which kept domestic melt production low between 2022-2024.

This pattern remained in place during this first half of 2025 – that is, during this time of strong melt output at home – but even here the interpretation remains unclear: has there been real demand for consumption at home, pushed upwards by the actual and potential additional tariffs promised by the current administration? Or has this been re-stocking by distributors, keen to replenish material at this time of low stainless prices from both home and imported sources? Import data shows that basic grades are coming in increasingly from Indonesia, India, Vietnam and Taiwan, while material from the higher cost centres of Japan, South Korea, France and Germany have flatlined. This dovetails well with our own expectation that US melters, much like those in other developed countries, are having to move away from standard grades in favour higher value niche products.

And it is largely for these reasons that we consider any sustained expansion above 2 Mtpa of melt production in the US to be unlikely over the long term. Furthermore, the high scrap ratio of 75-80% and a stable preference for austenitics (70% of total melt production) is likely to keep primary nickel consumption in stainless stable also, at 25-30 ktpa. The move towards higher stainless steel grades might imply slightly rising nickel content. But, as we are seeing in Japan at least, we might also begin to see higher use of specialist ferritics, where the inclusion of other alloying elements can in some applications do the job that nickel does but at lower cost.

Outside stainless, but equally suffering from the effects of the current administration's trade- and subsidy-related policies, the renewable energy businesses is in a holding pattern. At the "downstream" end of the EV segment, the US\$7,500 sales subsidy on certain vehicles ends this month, and so while we expect to see strong EV sales through the end of

Q3 2025, we anticipate considerable decline thereafter. Further “upstream”, the August “raid” by immigration officials at the Hyundai-LGES battery gigafactory currently under construction in Georgia will not just delay completion of that plant but is sending shockwaves through the entire EV raw materials supply chain, most of which involves a mix of domestic and foreign investment but is overwhelmingly sustained by foreign technology. There is considerable doubt over the direction that this sector will take for the near term.

That doubt has extended further north into Canada – but for a different reason. Ontario and Quebec were the provinces leading the renewables charge, from lithium mining and processing through to sulphates processing, cell manufacturing and battery pack assembly. Six months ago, several projects were underway, not entirely in an integrated fashion but nevertheless putting Canada on the “ex-China” map regarding electrification. But virtually all of this has been put on hold for the foreseeable future as cratering lithium prices and indecision by OEMs regarding the North American EV landscape have contributed to an unwillingness to complete financing. Projects currently on the drawing boards might still get built, but we expect lower capacities and slower ramp-ups will now be the norm.

Nickel in battery precursors

Global nickel consumption in battery precursors is forecast to reach 560 kt in 2025, with China, at 449 kt, remaining the cornerstone of global demand. Yet China’s demand dynamics are undergoing two pivotal shifts that will shape the market over the medium to long term.

First, the shift toward LFP chemistries has slowed new pCAM capacity additions. Near-term headwinds are evident in recent project deferrals and suspensions, including Dowstone’s integrated 100 kt pCAM with 50 kt nickel sulphate facility, and Ronbay Technology’s 30 kt pCAM expansion in Zhejiang province. Notably, these project adjustments are not large enough to offset broader capacity expansion plans which allows us to maintain our forecast of a 10.6% CAGR for China’s pCAM production over the next ten years.

Second, China is in the transition period from mid-nickel to high-nickel pCAM chemistries. Mid-nickel chemistries such as NCM 523 and 622 remain dominant, accounting for over 50% of domestic nickel-based pCAM production in 2025. This is due to their balance of energy density and cost, making them ideal for mass-market EVs. We expect their market share to grow further in the near term.

Over the long term, however, high nickel and ultra-high-nickel pCAMs will accelerate, driven by the commercialisation of solid-state batteries that require higher nickel content for performance. Leading producers are already advancing capabilities to meet this demand: GEM is collaborating with automakers to develop ultra-high-nickel pCAMs ($\text{Ni} \geq 9.5$, $\text{Co}+\text{Mn} \leq 0.5$) for solid-state batteries (targeting mass production in 2026) and has secured overseas automaker approval for its NCM 911 product; it is also retrofitting low-nickel lines to high-nickel capacity. CNGR, the world’s largest pCAM producer, noted in its H1 2025 report that it has built an ultra-high-nickel lab and aims to commercialise NCM 955 by 2027 to match solid-state battery requirements. Higher nickel content in these products will further increase per-unit nickel demand. Therefore, we expect nickel consumption in battery pCAMs to also increase with a CAGR of 10.6% over the next decade.

Ex-China, nickel consumption is constrained by scarce local pCAM capacity. North America and Europe face local pCAM capacity gaps, worsened by our downgrade of key projects this quarter, such as Redwood Materials’ 100 ktpa cathode active materials (CAM) facility. South Korean producers are ramping up local capacity to supply North America and capture US EV subsidies, but their expansion pace will be slowed by a critical barrier: US subsidy rules require “non-China” upstream materials (including MHP and nickel sulphate) to qualify. Since global MHP and nickel sulfate production remains largely dominated by Chinese companies, South Korean producers could struggle to secure compliant feedstocks, which will limit the speed and scale of their pCAM capacity additions. This feedstock bottleneck, in turn, will ultimately temper ex-China pCAM supply growth and cap associated nickel consumption expansion.

Is electrification on pause?

A decade on from its beginnings on a global scale, we are witnessing the first “crisis” in what has been dubbed the market of nickel’s salvation. Nickel demand in pCAMs for renewable batteries, largely destined for EVs and energy storage systems (ESS), rose by “only” 6.1% in 2024 to 519 kt. This year we are currently forecasting 8% year-on-year growth, globally to 560 kt. Even though that sounds like an acceptable improvement, we should point out that this figure is heavily dependent on growth projects in China ramping up in the second half of the year. In H1, available data for China point to a contraction of 4% compared with H1 2024, and so this reliance on project construction completion and ramp-up is a risk to our current year reality when data become available in early 2026.

Equally, for the same reason that we show a strong H2 2025, next year also looks strong, with 14% year-on-year growth in China and 15% globally. This implies almost 20% growth in 2026 from production facilities outside China. Indonesia and Morocco will be the main drivers here – and so our “ex-China” forecast is still reliant largely on China’s willingness to finance such projects. But we should also remind ourselves that in tonnage terms, that growth is only about 20 kt on the low 2025 base of 110 kt.

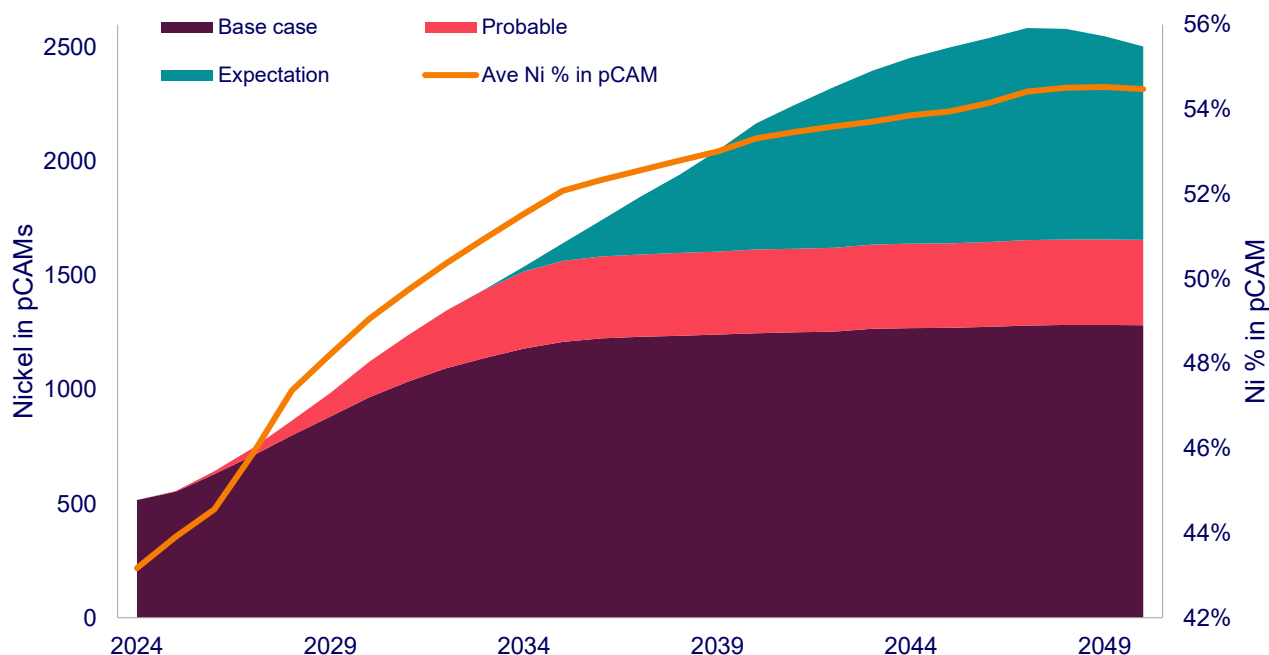
That the members of the EU cannot find a common path to an electrified future right now – and that the US is also in reverse gear on several renewables initiatives put in place by the previous administration – makes nickel demand from pCAMs an even more Asia-focused story. China’s investments at home and internationally tighten the nation’s grip on the world’s electrified future. But it also exacerbates the risks to our forecasts for the long term.

While our near to mid-term forecast of nickel consumption in battery chemicals is determined from our knowledge of existing pCAM plants and projects, the long-term outlook is guided by forecasts of EV sales dovetailed with gigafactory projections from other Wood Mackenzie teams. Inevitably, from the late 2030s there is an ever-escalating difference between these forecasts and the estimates based on existing pCAM facilities because we will need new facilities to keep up with that demand. But it is too soon in 2025 for companies to be planning new capacity in the 2040s.

In 2035, that “expected” level of nickel demand from unidentified pCAM/CAM plants stands at 77 kt, representing 4.7% of our global nickel demand of 1.63 Mt. By the mid-2040s, this figure rises to 800-850 kt, implying that one third of our forecast nickel demand in precursors will come from those pCAM sources that do not currently exist.

Therefore, to achieve our long-term forecast of nickel demand in batteries there is an intrinsic assumption that there will be new pCAM production from an as yet unknown number of plants. We make the further assumption that this unknown source of pCAM production will be Chinese and so the main driver of long-term demand for nickel in pCAMs continues to be China.

Risks to our global nickel use in precursors

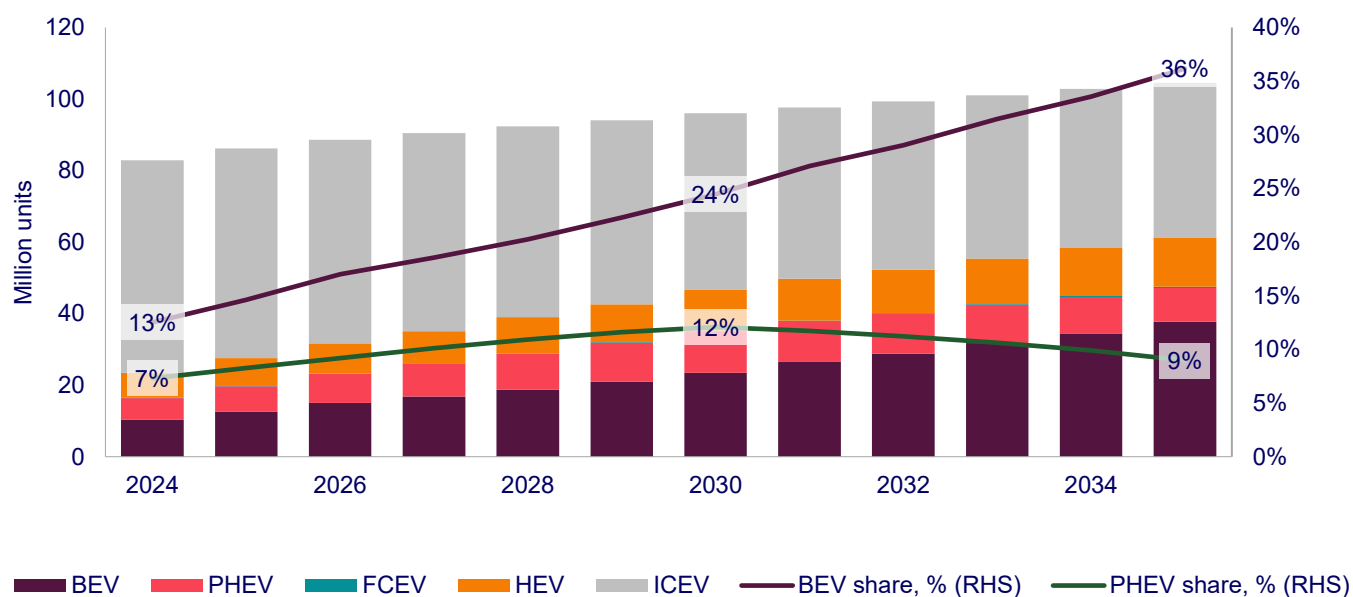


Source: Wood Mackenzie; ZLJ Steel

A second risk, as shown in the graph above, is the sharp rise in the percent ratio of nickel contained in pCAMs. Beyond the problematic debate about how and why the average chemistry of the whole industry should evolve over the next two decades lies the inescapable fact that there is a lack of consistent data available on the product mix of individual plants and companies themselves, and how this can change from quarter to quarter. It might be that the discussion about why chemistry types evolve over the long-term, as well as the threats posed by non-nickel-containing chemistries, will change this ratio in subsequent reports, potentially reducing the requirement for nickel in a similar way to how cobalt’s trajectory has changed over the last few years.

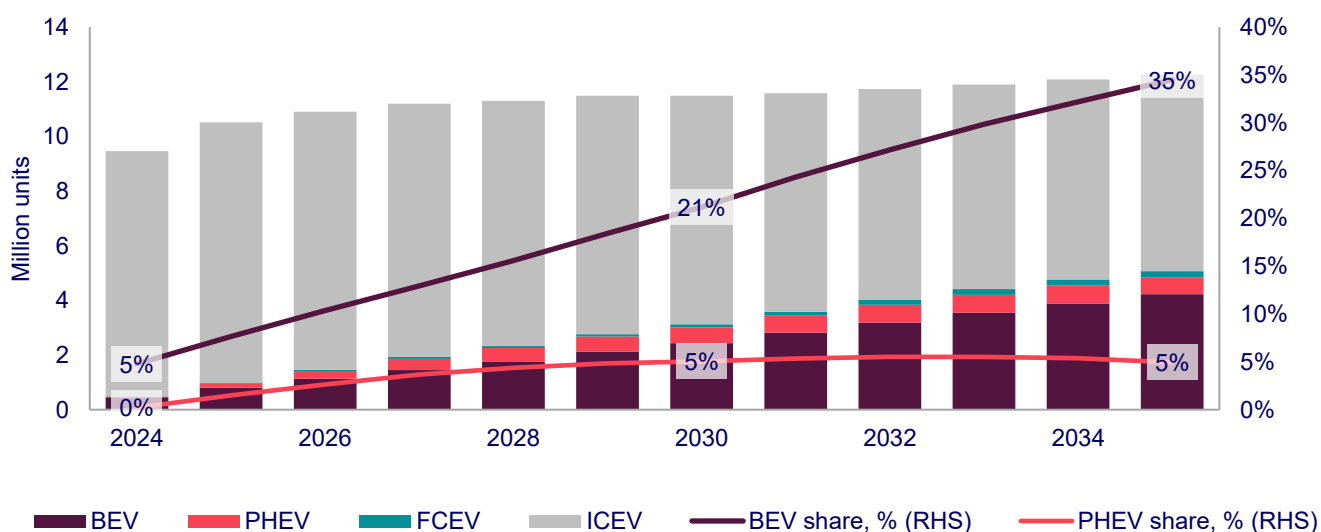
A third risk comes from the potential for EV sector contraction in China. Last month our colleagues in China issued an Insight focusing on “anti-involution”: [Value over volume: anti-involution to revitalise China's EV and battery supply chain](#). In short, the many ways that Beijing will encourage all industrial sectors to rid themselves of heavy subsidies and bloated competition will probably lead to a reduction of players in the electrification field, starting with the manufacturers of EVs themselves and moving throughout the chain, down to nickel chemicals producers and precursor consumers. Although in the longer term these “policies” are likely to allow true low-cost players to win, in the short- to medium-term such contraction would lead to lower demand. Beijing cannot enforce these anti-involution measures on these privately-owned businesses in the same way that it did with state-owned steelmakers back in 2015, but the effects of support programmes like subsidy withdrawal and imposing new barriers, such as raising quality standards, will take their toll.

Passenger vehicle sales and EV share



Source: Wood Mackenzie

Commercial vehicle sales and EV share



Source: Wood Mackenzie

Supply

Overview

We expect moderate growth in nickel supply over the next 10 years. Refined nickel production is forecast to increase by 1.0% a year through to 2035, reaching 4.34 Mt by that time (before adjustments). Over the same period mined nickel production increases by 0.9% a year, from 4.14 Mt in 2025 to 4.52 Mt in 2035.

Regional nickel mine production capability

	2025	2026	2027	2028	2030	2035	2040	2050	CAGR '25-2035	Prod'n kt '25-2035	CAGR '25-2050
Indonesia	2647	2817	3003	3104	3133	3133	3133	3133	1.7%	487	0.7%
Philippines	396	305	305	305	305	305	305	273	-2.6%	-91	-1.5%
Russian Fedn	191	201	214	216	217	217	217	217	1.3%	26	0.5%
Canada	135	155	163	167	146	136	52	0	0.1%	1	-100.0%
Australia	57	59	48	40	40	40	40	0	-3.5%	-17	-100.0%
N. Caledonia	146	191	207	213	213	213	213	213	3.8%	66	1.5%
Others	562	597	606	608	605	470	463	311	-1.8%	-92	-2.3%
Global Total	4136	4325	4547	4653	4659	4515	4424	4147	0.9%	379	-2.3%

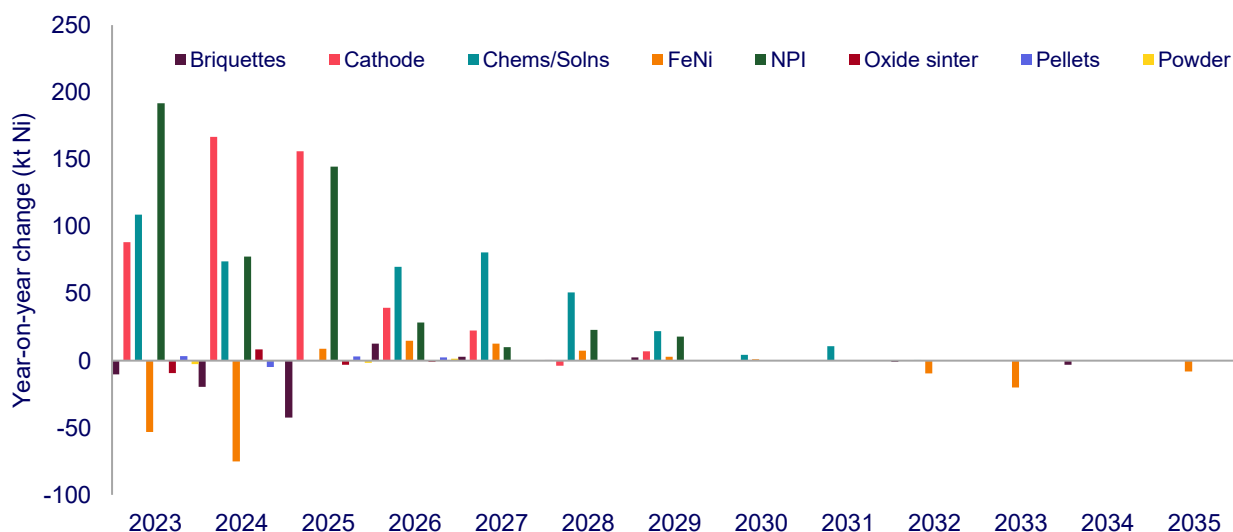
Source: Wood Mackenzie

Regional nickel refinery production capability

	2025	2026	2027	2028	2030	2035	2040	2050	CAGR '25-2035	Prod'n kt '25-2035	CAGR '25-2050
China	1208	1247	1246	1255	1243	1243	1243	1243	0.3%	35	0.1%
Japan	108	107	107	107	107	107	107	107	-0.1%	-1	0.0%
Indonesia	1848	1945	2013	2053	2100	2100	2100	2070	1.3%	252	0.5%
Norway	100	100	100	100	92	92	92	92	-0.8%	-8	-0.3%
Russian Fedn	124	129	139	141	142	142	142	142	1.4%	18	0.6%
Canada	123	129	131	131	131	130	87	57	0.6%	7	-3.0%
Australia	35	37	37	37	37	37	37	0	0.6%	2	-100.0%
Others	389	409	459	485	516	486	486	424	2.2%	97	0.3%
Global Total	3935	4103	4232	4310	4368	4338	4295	4136	1.0%	403	0.2%

Source: Wood Mackenzie

Nickel production by product type



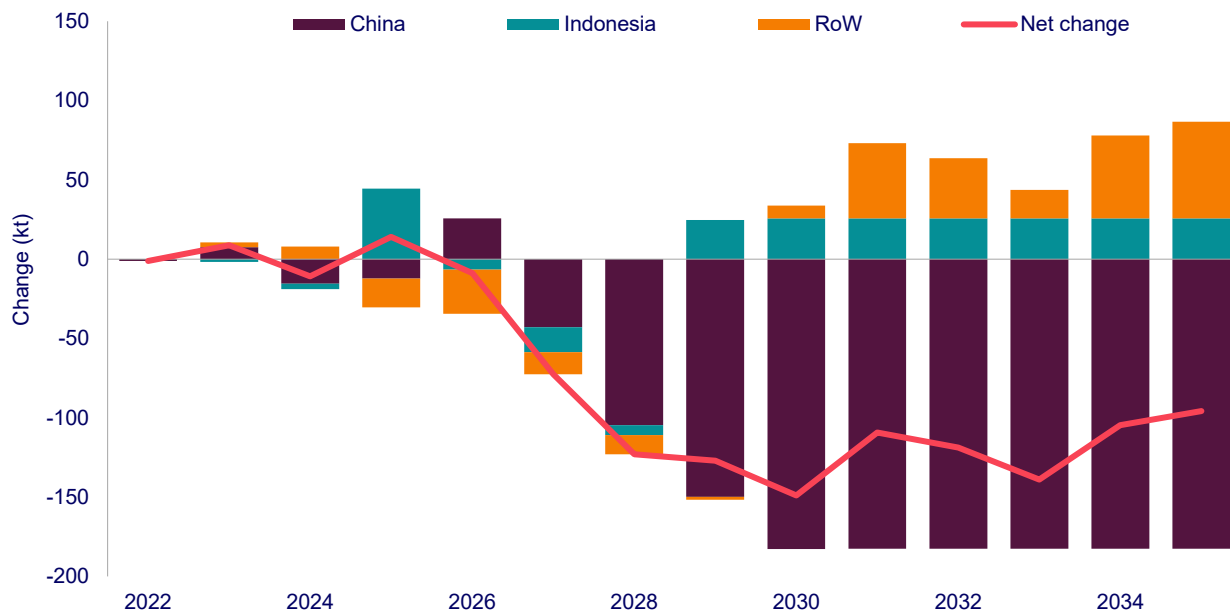
Source: Wood Mackenzie

While refined output shows annual growth of 1.0% over the forecast period, the overwhelming net increase in primary nickel product over that period is in nickel sulphate, which accounts for 44% of the supply increase, followed by nickel briquettes and cathode production, at 13% and 7% respectively.

Changes since the last quarter

The main changes this quarter are a cut in Chinese chemicals output over the mid-to longer-term. This has been brought about by a slowdown in HPAL and matte facility build out in Indonesia, itself a response to an increase in the proportion of LFP in the battery mix over the longer-term as well as a slowdown in EV uptake forecasts.

Main changes in refined nickel production since the previous quarter



Source: Wood Mackenzie

In Indonesia the increase in output from chemicals and cathode for 2025 is due to the commissioning of Qingmeibang (30 ktpa nickel cathode), QMB and Huaxiang (30 ktpa and 50 ktpa, respectively, nickel in sulphate).

Changes in the RoW are primarily life extensions of Voiseys Bay and Vale Sudbury, as well as the addition of the Onsan refinery in Korea.

The net change is an average cut of more than 100 ktpa from 2026 through to 2035.

The supply landscape

Africa – the new nickel sulphide frontier?

The nickel price decline over the past two years has resulted in an 8% fall in nickel-in-sulphide-concentrate production since 2023. Much of this capacity loss has been in Australia. Combined with the closure of Ravensthorpe in the laterite sector, there is now a possibility that Wales will produce more finished nickel than Australia in 2025!

While one continent “closes”, another is re-launching: Africa. And while there is some Chinese investment involved, a lot of this potential is being driven by the west.

Kabanga (Tanzania), Selkirk and Selebi (Botswana) have been actively progressing for several years now. The revisit to the possible reactivation of Nkomati (South Africa) got under way in 2023. And, in the last few months, what were probably considered dormant projects at best, Ntaka Hill (Tanzania) and Tati (Botswana), have returned to the table. If reports are to be believed, a number of these projects will be in production by next year! Already there, apparently, is Zhongzhou Mining Co. Ltd. which has started construction of a nickel/copper processing facility in Dodoma (Tanzania).

The overriding narrative is that Africa appears to be being widely seen as having nickel sulphide development potential. Challenges remain with infrastructure, transport and in some instances, resource grades. But there is a momentum

building, and a willingness by host nations for job creation. A region to watch in the coming decade.

Kabanga

The Kabanga project is in northwestern Tanzania, south of Lake Victoria near the Burundi border, and is regarded as one of the world's largest undeveloped greenfield nickel sulphide deposits. Current proven and probable reserve stand at 43.9 Mt grading 1.98% Ni and 0.27% Cu. There is an additional inferred resource of 13.5 Mt grading 2.08% Ni and 0.28% Cu.

Current owners, Lifezone Metals, initially proposed a two-phase development plan with 3.4 Mtpa aggregate capacity, including both mine-mill operations and a downstream hydrometallurgical refinery at Kahama. The June 2025 initial assessment maintained this integrated approach, featuring a 22-year mine plan transitioning from concentrate production to battery-grade materials at the refinery after five years. However, the July 2025 feasibility study reduced scope, removing the Kahama refinery and focusing on an 18-year operation with a capex of US\$943 million.

The project envisaged would be constructed over a 32-month period up to first concentrate production. Steady state will produce ca. 350 ktpa of nickel concentrate grading 17.7% Ni (for around 62 ktpa nickel, with an average of 50 ktpa).

Previous owners, Glencore and Barrick, had similarly focused on a concentrate-only production route, with the product being shipped to Canada for processing. The remote location of Kabanga meant that freight was around US\$200/t concentrate which always made the project economics difficult.

Earliest production date in our view would be 2029 assuming funding was secured by the end of this year.

Selkirk & Selebi

Having operated from 1989 as an underground mine, Selkirk ceased production in 2002. Then shareholder, LionOre, identified a 230.6 Mt indicated resource grading 0.24% Ni and 0.21% Cu and evaluated its development as a large-scale open pit.

The purchase of LionOre by Norilsk Nickel (Nornickel since 2016) in 2007 and subsequent sale to BCL Limited resulted in protracted litigation because of non-payment of dues. Ultimately the issues were resolved and NexMetals acquired Selebi and Selkirk in 2022. There is a smelter at BCL (the largest nickel flash furnace in the world) which was placed on care and maintenance in 2015, but this and its infrastructure were excluded from the mine's agreement.

NexMetals provided an initial mineral resource estimate in August 2024 which stated an indicated resource of 3 Mt grading 0.98% Ni and 0.9% Cu, plus an additional inferred resource of 24.7 Mt grading 0.92% Ni and 1.5% Cu. Selkirk has an inferred resource of 44.2 Mt grading 0.24% Ni and 0.3% Cu.

While active, production from either of these deposits is some years away.

Nkomati

Nkomati mine is located in the Machadodorp area of South Africa's Mpumalanga province, approximately 300 km east of Johannesburg. The mine contains substantial disseminated copper-nickel sulphide ore resources across several ore bodies, including a high-grade solid sulphide deposit and a peridotite chromite zone with lower nickel but higher chrome content. As of 31 December, 2020, total resources were estimated at 214.8 Mt, grading 0.36% Ni and 0.14% Cu.

Open-pit operations ceased in March 2021, when joint venture partners Nornickel and Africa Rainbow Minerals (ARM) placed the mine on care and maintenance.

In July 2025, Nornickel completed the divestment of its 50% stake in the Nkomati Joint Venture through its subsidiary, Norilsk Nickel Africa Proprietary Limited, with ARM acquiring full ownership of the mine.

Under the agreement, ARM assumed all environmental liabilities and Nornickel's proportionate share of obligations related to the mine's assets. This full ownership gives ARM complete control over future development decisions and potential operational restart.

The mine is ostensibly being managed towards closure. However, ARM continues to explore the possibility of restarting the underground mine should platinum group metal and/or nickel prices prove favourable. The company also views the mine as having a "low-carbon" potential as, ultimately, a preferred source of nickel into the battery sector. Historically

the mine produced around 6-8 ktpa nickel in concentrate over the last five years of operation.

Given our nickel price outlook, any restart is unlikely before 2030.

Ntaka Hill

Ntaka Hill is a nickel sulphide deposit in southern Tanzania's Neoproterozoic Mozambique belt. Australian Stock Exchange-listed Indiana Resources controlled the asset for more than 20 years. The deposit has a mineral resource of 56.2 Mt grading 0.63% Ni and 0.14% Cu.

A protracted legal dispute began in January 2018 when Tanzania's government expropriated the project licence. The government abolished the "Retention Licence" without offering replacement classification. Indiana Resources responded by launching a legal challenge through international arbitration. The International Centre for Settlement of Investment Disputes (ICSID) awarded Indiana Resources over US\$109 million in July 2023.

Indiana Resources reached a landmark settlement with Tanzania in 2024 over the unlawful expropriation. The Tanzanian government agreed to pay US\$90 million to resolve the dispute. This settlement represents 82.5% of the original US\$109 million ICSID award.

Tanzania's State Mining Corporation (STAMICO) announced plans in August 2025 to establish a nickel and copper processing plant at Ntaka. The facility will operate in partnership with China's Ningbo Shuangneng Group. The plant will process feed from the Ntaka Hill site using "advanced" Chinese technology. Twenty-four of 38 shipping containers containing plant equipment have arrived in Tanzania. Ningbo Shuangneng Group has confirmed readiness to finance and provide technology for immediate construction, pending geological verification completion.

Clearly this is a huge shift in the potential development of Ntaka Hill. However, what product and at what level of output may be produced is not known.

Tataki (formerly Tati)

Operations at the Tati Nickel mine in Botswana began with the Selkirk underground mine in 1989 and the Phoenix open pit in 1995. The mine changed hands from Norilsk to BCL in 2015, but operations ceased in H1 2016 due to significant losses.

The mine has been acquired and is being restarted by NIU Group, a London-based investment firm headed by Austrian investor Cevdet Caner. NIU plans to invest around US\$200 million in capital over a 10-year period. This includes US\$50 million over the initial 18 months to reach steady-state production. Production from Tati, now renamed Tataki, will ultimately combine open pit mining with hydrometallurgical processing. Operations will process approximately 40 Mt of run-of-mine material from the open pit and reprocess over 80 Mt of surface stockpiles. Estimated production includes 170 kt of nickel hydroxide, 100kt of copper cathode and 7 kt of cobalt hydroxide over the mine life (equivalent to ca. 5 ktpa Ni, 10 ktpa Cu and 250 tpa Co).

Operational re-start is apparently already underway, although what the initial product will be has not been stated.

Dodoma processing facility

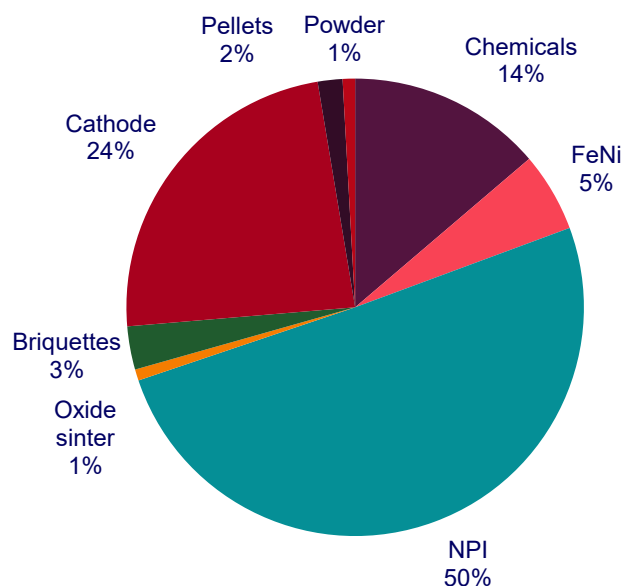
In August 2025, Tanzania's Minerals Minister, Anthony Mavunde, laid the foundation stone for a Zhongzhou Mining nickel-copper refining/processing plant in Zamahero Village, Bahi District (Dodoma), with an estimated capex of over US\$15 million.

The project will use a "two-stage countercurrent leaching-extraction-electrodeposition" hydrometallurgical process. Nickel and copper recovery rates are put at better than 92% and 95%, respectively. Final products will be 12 kt of nickel sulphate (ca. 2.7 kt nickel) and 8 kt of copper sulphate (ca. 2 kt copper contained). It is also proposed to utilise the tailings, mixed with cement, to produce bricks with a view to achieving a zero-waste discharge.

We assume the feed would be nickel concentrate, but this is unclear.

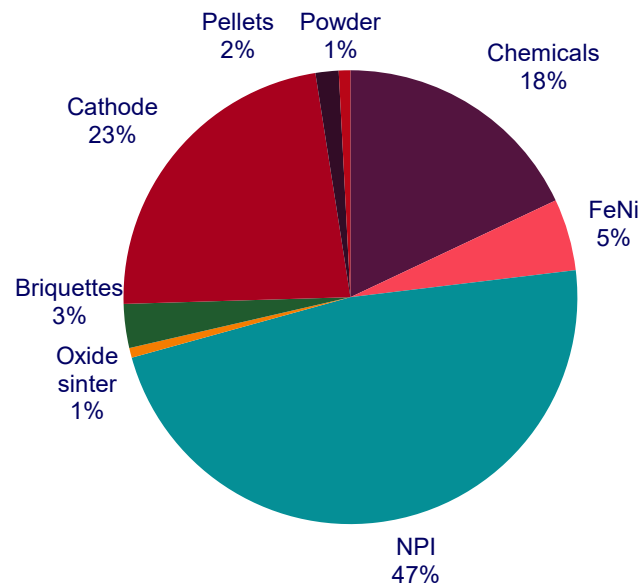
Nickel production by type

Nickel production by product 2025 (3.94 Mt)



Source: Wood Mackenzie

Nickel production by product 2035 (4.34 Mt)

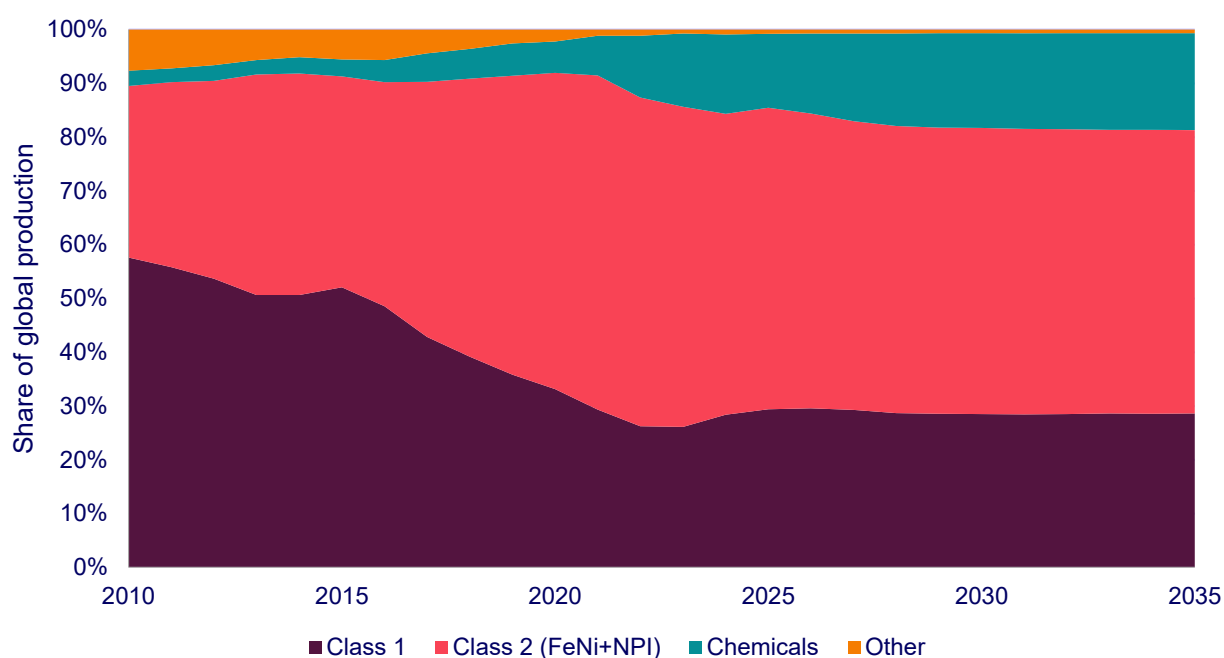


Source: Wood Mackenzie

Nickel production has moved away from the “traditional” Class I nickel to more market specific products – namely NPI and chemicals. Class I nickel slipped below 60% of the market in 2010, and the decline in share has continued unabated, to only 26% in 2023.

This is driven by market requirements. The largest nickel market of stainless steel needs FeNi and NPI, and much less Class I. The developing battery sector, while initially relying on Class I metal dissolution to make nickel sulphate, is now fed by nickel intermediates (matte and MHP) – removing almost entirely the requirement for metal dissolution (except in some cases outside China). So, it is not surprising that there has been very little development of new Class I-producing facilities as there has been no demand growth for Class I nickel units.

Refined production by class



However, more recently we have seen an increase in Class I production. This has been driven not by some new market or increase in demand, but by oversupply of nickel sulphate and a divergence in price between metal and sulphate in China. In addition, the LME opened its doors to fast-tracking new nickel brands to increase liquidity in the market. This combination has resulted in seven new brands being accepted as good delivery since July 2023 – representing around 175 kt of capacity. All the new brands are Chinese owned, with one produced in Indonesia.

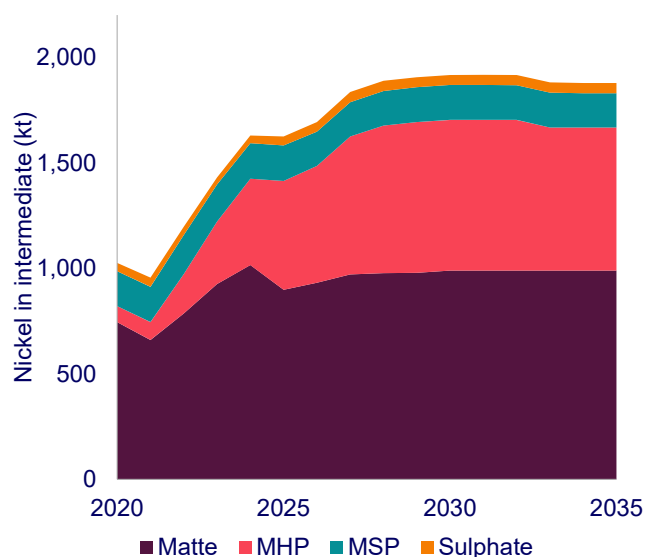
This switch to producing material for the market of last resort, the LME (and/or SHFE), has arrested the anticipated further decline in Class I metal as a proportion of global output. But with no increase in demand, the metal will end up on the exchanges, increasing visible stock levels and likely, therefore, to put downward pressure on price. However, the loss of western Class I production through the closure of Sandouville and Nickel West will mean that non-FEOC (foreign entity of concern) material availability will become increasingly tight.

It should also be noted that this new Chinese Class I output is swing production from sulphate. If battery demand returns strongly then sulphate demand will increase, which will mean nickel metal production gets switched out.

Intermediates markets

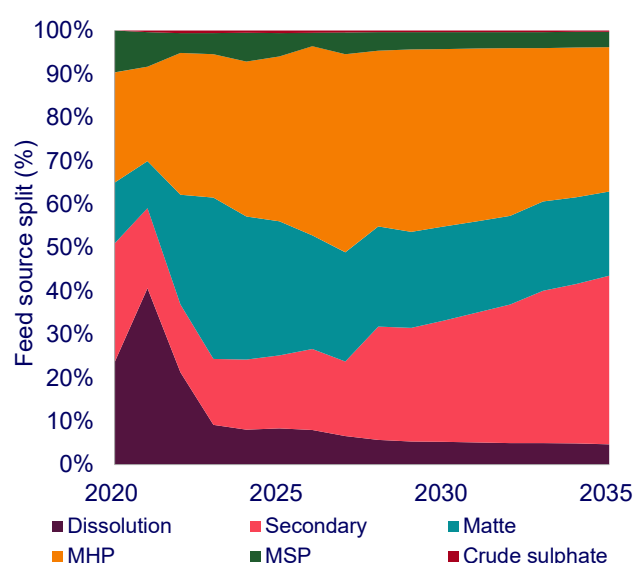
The growth in nickel demand for nickel sulphate for EV batteries is currently the driving force behind the rapid development of intermediate production, largely in Indonesia. The preference for the use of intermediates rather than finished nickel dissolution (for example) is driven by the lower associated costs, as well as the ability to also provide cobalt units. MHP is preferable to matte, as it does not require pressure to get nickel and cobalt into solution. For that reason, the capital requirement for processing MHP is lower compared with that for matte on the same scale.

Nickel in intermediates supply (kt)



Source: Wood Mackenzie

Nickel in intermediates for chemicals – by share



Source: Wood Mackenzie

Currently around 70% of nickel sulphate is produced using matte and MHP feed, compared with around 40% in 2020. The use of MHP has grown from just 9% in 2015 to 36%, currently, and is expected to remain at this level through to 2034. Absolute tonnage of nickel in MHP usage rises from 222 kt currently to 345 kt by 2035. The use of recycled materials is also growing. We currently forecast a 300% increase in the production of nickel sulphate from recycled sources over the same time frame, from 98 kt nickel to 405 kt by 2035.

A development during 2023 and still ongoing is the conversion of nickel sulphate to nickel metal. Due to the huge growth in intermediates production for sulphate for EV batteries and the subsequent slow-down in demand from that sector, producers have turned to converting excess sulphate to metal. We forecast nickel cathode production reaching almost 1 Mt by 2035, compared with 777 kt in 2024. However, as noted above, the production of metal is interchangeable with the delivery of sulphate and could go down should battery demand rise and additional sulphate be required.

Intermediate payabilities

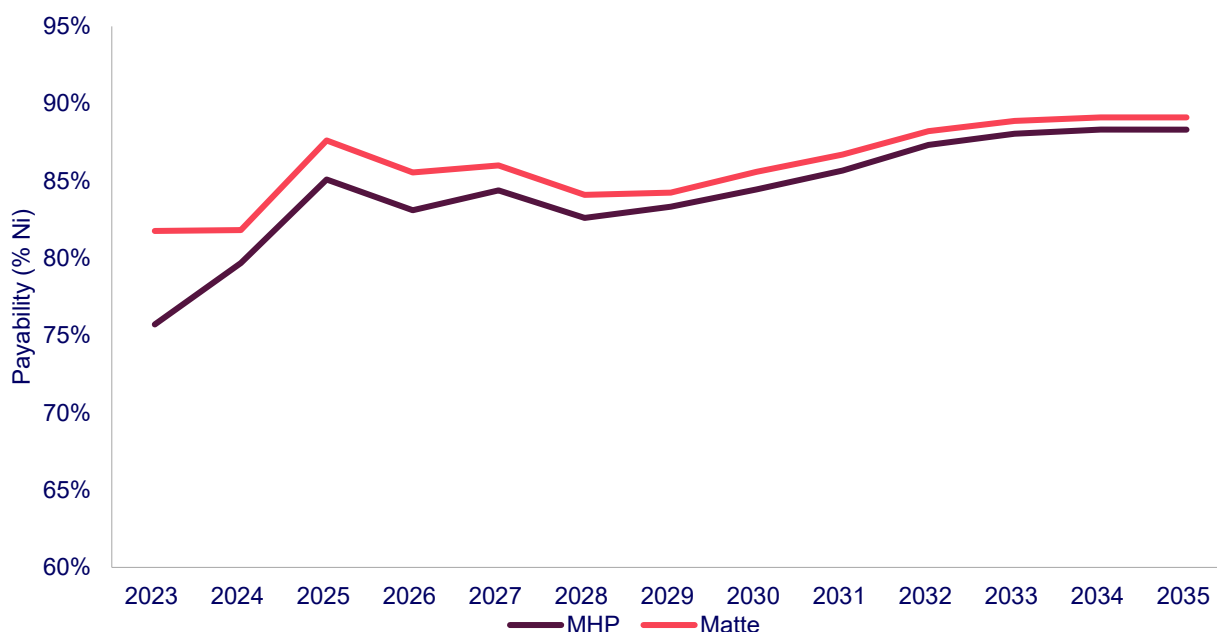
With the increased availability of nickel intermediates, and an increasing number of would-be producers, especially of MHP for the battery market, there has been an increased requirement for estimates on nickel-in-MHP and -matte payability forecasts.

As with nickel sulphate, this is a market dominated by Chinese purchasing power and Chinese-controlled production, and as such our forecast are indications based on a CIF China basis.

Our methodology is based on both our nickel and nickel sulphate price differentials, the costs of processing matte and MHP to sulphate and the potential to produce metal, at an additional cost, all combined with an assumed margin. Using these parameters, our current payability forecasts for nickel-in-matte and MHP are shown in the chart below.

Essentially, payabilities for matte and MHP fall in line with our declining nickel price forecast before increasing from 2029 through to the end of the forecast period, reaching 89% and 88%, respectively.

Payabilities for nickel in matte and MHP



Source: Wood Mackenzie

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